CIFAR-10 / MLP
Adam, $\eta = 0.001$; peak 703

CIFAR-10 / ResNet-32
Adam, $\eta = 0.003$; peak 422

WikiText-2 / Transformer
Adam, $\eta = 0.003$; peak 794

(a) (b) (c)

Training loss

(d) (e) (f)

$L_t - L_{t-1}$

(g) (h) (i)

$\hat{\eta\kappa}_t$ (raw gradients)

(j) (k) (l)

ρ_t (raw gradients)

(m) (n) (o)

ρ_t (momentum)

(p) (q) (r)

ρ_t (Adam)

(s) (t) (u)

$\|g_t\|$

(v) (w) (x)

$\|m_t\|$

(y) (z) (aa)

R_t (Adam)

Training state t

All observations unsmoothed. Light dashed: before loss crosses starting value; dark dashed: peak.

g_t, m_t and adaptive diagnostics are aligned to state t, before the next parameter update.

R_t retains actual updates through state t. Windows end at the first loss below their starting value.